

SENTHIL VELAN P

ENTERPRISE AI & DATA TRANSFORMATION DIRECTOR

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Enterprise AI & Data Delivery Executive with over **20 years of experience** leading large-scale AI, Data Engineering, Analytics, and Cloud transformation programs across Telecom, BFSI, Retail, Healthcare, and enterprise digital transformation engagements. Proven expertise delivering enterprise AI initiatives across the complete lifecycle—**Strategy, Architecture, Design, Build, Deploy, MLOps, Production Operations, and Enterprise Scale Adoption**. Extensive experience managing global engineering organizations of **500+ professionals**, transformation portfolios exceeding **\$29M**, executive stakeholder relationships, AI/Data operating models, and full P&L accountability. Strong background in Data Engineering, Data Governance, AI Program Governance, GenAI industrialization, and enterprise-scale AI platform modernization.

CORE COMPETENCIES

Data Engineering & Platform Governance	Data Engineering, Data Architecture Oversight, Data Platform Governance, Data Quality Frameworks, Master Data Management (MDM), Apache Kafka, Google BigQuery, Cloud Data Lakehouses (Snowflake, Databricks), Enterprise Analytics.
Cloud-Native Modernization	Monolith-to-Microservices Migration, Elastic Containerization (Kubernetes, GKE), DevOps/DevSecOps Pipeline Engineering, SRE TechOps, Hybrid Cloud Solutions (AWS, GCP).
Executive Delivery & program management	Portfolio delivery management(\$29M+), Revenue & Margin Optimization (Consistently Achieving 25%–30% RGM), RFP & Bid Architecture, Global Capability Center (GCC) Governance.
Enterprise AI & Data Leadership	Enterprise AI Strategy, AI Program Delivery, Agentic AI Frameworks, Multi-Agent Orchestration, Retrieval-Augmented Generation (RAG) , LLM Orchestration, LLM Guardrails, Responsible AI, Enterprise AI Adoption Frameworks.
AI Governance & MLOps	Model Lifecycle Management, MLOps Pipelines, Production AI Operations, AI Governance, Model Monitoring & Drift Detection, Human-in-the-Loop Validation , AI Risk Management, AI Ethics & Compliance.
Enterprise Governance & Compliance	Executive Stakeholder Management, Vendor & Partner Management, Resource & Pyramid Management, RAID Governance, Risk & Issue Management, GDPR & DPDP Compliance, Enterprise Data Privacy , Go-Live Readiness & Executive Reporting.

PROFESSIONAL EXPERIENCE

Director – Technology, AI & Data Transformation Brillio	2024 – Present
Chennai, Tamil Nadu	
Enterprise AI transformation programs: Led across the complete lifecycle—from AI strategy definition, solution architecture, data platform readiness, model engineering, deployment, production stabilization, and enterprise-scale adoption.	

- **Production-Grade Agentic AI:** Pioneered the design and engineering deployment of autonomous Agentic AI models and multi-agent workflows within client systems to automate complex transactional operations.
- **Contextual GenAI Solutions:** Spearheaded the development of a secure, LLM-powered chatbot utilizing Retrieval-Augmented Generation (RAG) to serve NRB engineers, accelerating technical knowledge retrieval and cutting resolution cycles.
- **GenAI-Driven UI/UX Transformation:** Architected intelligent, adaptive user interfaces powered by Generative AI to deliver hyper-personalized user journeys, dynamic content layout adjustments, and conversational UI elements that significantly enhanced digital engagement.
- **Predictive Churn Analytics:** Designed real-time predictive ML pipelines utilizing Google BigQuery and Python to deliver actionable churn analytics and hyper-personalized portfolio recommendations, enhancing subscriber retention metrics.
- **P&L & Portfolio Optimization:** Directed 500+ strong cross-functional engineering organization. Structured commercial delivery models that sustained a 30% average project profit margin and achieved a 15% reduction in client operational costs through GenAI-enabled engineering automation.
- **RFP and Solution Consulting:** Partnered with sales and enterprise architecture units to deliver end-to-end responses for high-stakes RFPs, explicitly framing GenAI delivery methodologies to capture new market opportunities.
- **AI Delivery Governance & MLOps:** Established enterprise AI delivery governance and MLOps frameworks covering model lifecycle management, deployment automation, monitoring, drift detection, and production operations while driving data governance, data quality, MDM, and Responsible AI practices through LLM guardrails, access controls, human-in-the-loop validation, and AI risk management across enterprise GenAI initiatives

Account Head – Data Transformation | Incedo

2022 – 2023

Chennai, Tamil Nadu

- **Financial Risk & Predictive Telemetry:** Directed the development of supervised machine learning models (Regression, Random Forests) to forecast real-time credit risk behavior, identify fraudulent transaction spikes, and isolate security anomalies across high-throughput BFSI systems.
- **High-Performance Data Mediation:** Engineered ultra-low-latency data mediation infrastructures leveraging Apache Kafka and Bloom Filter caching, enabling real-time stream optimization to feed downstream ML trading and fraud-detection pipelines.
- **Observability Architecture:** Directed the rollout of comprehensive network and system performance monitoring systems utilizing Prometheus, Grafana, and the ELK Stack, integrating ML-driven threshold alerting to maintain strict SLA metrics for financial processing hubs.
- **Interactive Data Dashboards:** Oversaw the end-to-end design of real-time planning and scheduling platforms for FTTH (Fiber to the Home) deployment teams, engineered with custom ReactJS/NodeJS micro-frontends and driven by predictive optimization algorithms.
- **AI Delivery & Financial Governance:** Served as executive advisor for AI and Data transformation engagements, aligning technology delivery with financial KPIs, enterprise risk management, regulatory compliance, and long-term digital transformation roadmaps.

Senior Delivery Manager & Emerging Tech Solutions Head | Proapt

2016 – 2022

Europe

- **Legacy Core Modernization:** Orchestrated a massive monolith-to-microservices migration for a core legacy Order Management system, replacing brittle Pro*C/SQL frameworks with containerized Java microservices running natively on Google Kubernetes Engine (GKE) and AWS.
- **Classic ML & Telecom Analytics:** Conceptualized and led the deployment of classical machine learning algorithms (K-Means clustering, Logistic Regression) for customer segmentation and predictive billing pattern analysis, identifying revenue leakage trends.
- **Emerging Tech Commercialization:** Directed the architectural transition of legacy billing systems into distributed cloud environments, consistently driving 30% Revenue Gross Margin (RGM) optimizations across portfolios.
- **Alternate Engagement Models:** Conceptualized and executed alternative agile delivery frameworks that maximized team utilization and velocity while handling cross-border stakeholder escalations.
- **Change Management Leadership:** Formed and guided high-performance engineering squads through critical data modernization phases, establishing automated CI/CD practices and secure DevOps pipelines.

Technical Project Manager | Prodapt

2014 – 2016

Chennai, Tamil Nadu

- **Smart Infrastructure Solutions:** Managed the end-to-end engineering of Smart City data frameworks, emergency response systems, and scalable enterprise HRMS ecosystems hosted on AWS using Java, Spring, Angular, and message brokers.

Application Development Architect | Cognizant Technology Solutions

2008 – 2014

Chennai, Tamil Nadu

- **Distributed Services & Integration:** Designed high-throughput trading architectures utilizing Spring Boot, Spring Batch, Hibernate, and asynchronous Message Brokers (Apache Kafka, ActiveMQ, RabbitMQ).
- **Cloud Infrastructure & Observability:** Transitioned legacy multi-tier web applications into elastic AWS clouds (EC2, S3, Lambda, CloudFormation) while embedding performance visibility via Kibana, Grafana, and New Relic.

Senior Lead – Application Development | Honeywell Technology Solutions

2004 – 2008

Bangalore, Karnataka

- **Industrial Automation Engineering:** Developed and led core software components for smart commercial HVAC systems and industrial automation platforms using enterprise Java.

EDUCATION

Master of Engineering (M.E.) in Computer Science

PSG College of Technology, Anna University, Coimbatore, Tamil Nadu | Mar 2005

EXECUTIVE CERTIFICATIONS & TRAINING

- Leadership with AI – Indian School of Business (ISB)
- Leadership Development Program – Emeritus
- Agile Scrum Product Owner / ScrumMaster – Scrum Alliance
- DevSecOps & Cloud Solutioning Certified – Prodapt
- Sun Certified Java Programmer (SCJP 5) – Sun Microsystems

TECHNOLOGY LANDSCAPE

- **AI & Orchestration:** GenAI, Agentic AI Workflows, Multi-Agent Systems, RAG Architecture, LLM Orchestration (LangChain/LlamaIndex), LLM Guardrails & Governance, Vector Databases, Python, Predictive Analytics.
- **Data & Streaming:** Apache Kafka, Snowflake, Databricks, Google BigQuery, Bloom Filter Caching, ELK Stack, ETL/ELT Integration.
- **Cloud & DevOps:** AWS, GCP, GKE (Google Kubernetes Engine), Docker, Microservices Architecture, Micro Frontends, DevSecOps Pipelines, Prometheus, Grafana, New Relic.
- **Enterprise Frameworks:** Spring Boot, Microservices, Node.js, ReactJS, Messaging Brokers (RabbitMQ, ActiveMQ), Legacy Pro*C / C++, SQL.